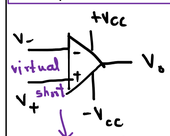
# 3. Opamp 2.0

17/03/25 9:15 PM

## Operational Amplifier

($R_{in} = \infty$, $R_o = 0$)




When in linear region

$A_d \to \infty$ (diffrential gain) [Av]

$A_c \to 0$ [common mode gain] [$V_{avg}$]

Slew rate: Max rate at which o/p of opamp can't change

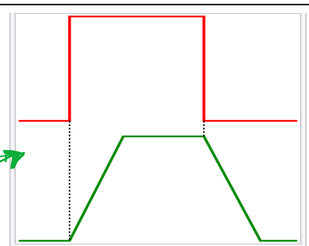
Inv, Non-inv opA etc. yk that.

# Saturatn of $V_o$ at $+V_{cc}$ or $-V_{cc}$

Ex. for some f. of input, o/p ✓ -op some it will show errors

CMRR → ∞

Slew Rate



slew rate effect on a square wave: red=desired output, green=actual output

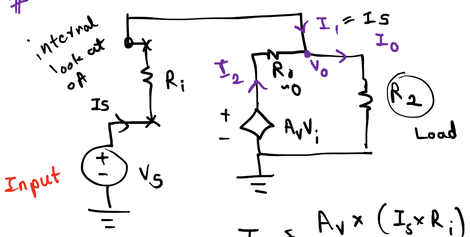
$A_o$ depends on freq. of inp. as well

$A_v = \frac{A_o}{1 + \frac{\omega}{\omega_c}}$, depends

Amplification of op amp depends on frequency of input

**Frequency Response:**
The gain of a real op-amp typically decreases as the input frequency increases, following a characteristic curve. This is because internal components, like capacitors, limit the op-amp's ability to amplify signals at very high frequencies.

#




$\frac{V_s}{I_s} = ?$ As seen from input $R_{in} \approx R_i(1 + A_v) \approx \infty$ high

$(R_o \to 0)$

$V_o = A_v V_i$

$I_2 = \frac{A_v \times (I_s \times R_i) - V_o}{R_o}$

$V_o = I_o R_2$

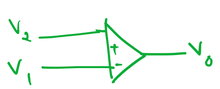
$\frac{V_s - V_o}{R_i} = I_s$

$R_{out} = \frac{R_o}{1 + A_v}$

TF??

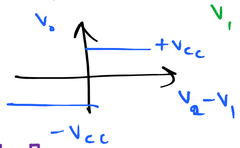
→ Offset Null

## Comparator: No feedback

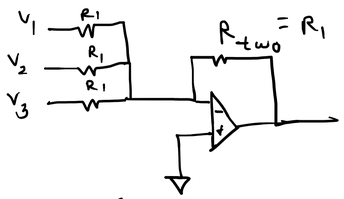



$V_2 > V_1$ : $V_o \to +V_{cc}$

$V_1 > V_2$ : $V_o \to -V_{cc}$

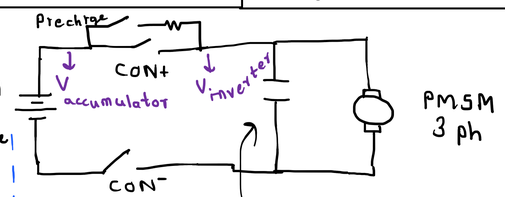



# Additn:




$V_o = -(V_1 + V_2 + V_3)$

1) CON⁻
2) precharge
3) 95% toh precharge & CON⁺




Inverter } DC to AC

95% ke baad hi precharge band krna hai

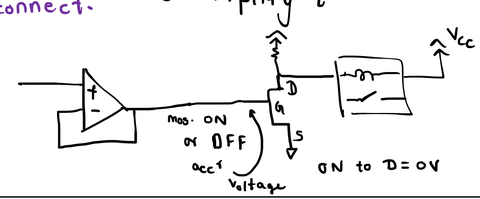
DC link capacitor: temp. pwr house

if V fluctuate hua toh it makes sure ki input shi se aaye.

opA ka o/p i is 50mA type, toh its not enough to start relay. So Amplify i

but buffer ka o/p directly nhi kr sakte connect.

Relay pwr ON ho isliye ek buffer lga do kuch LV source se




## TASK 101

# System critical signals (acc^t Rulebook)

↳ Error LEDs

We have to prove it works.

for ($I_s$ - $3s$) All ON

then only ERROR vale LEDs should stay ON

HOW?

Using opA, mosfets etc.

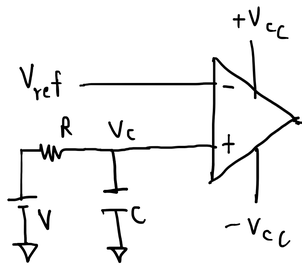
Seq of starting CAR

1) LVMS low Voltage
2) (high V) TSMS → master switch → tractive syst → doesn't ON HV, but HV won't be ON if TSMS is OFF
3) Some Logic HV Reset (ye dabao tb it'll ON HV)
4) Break dabate hue ignitn switch n stuff then if u acc^t; wheels turn

Comparator with RC ckt for $I_s$, $3s$ ON thingy

pmos

$t < 1$

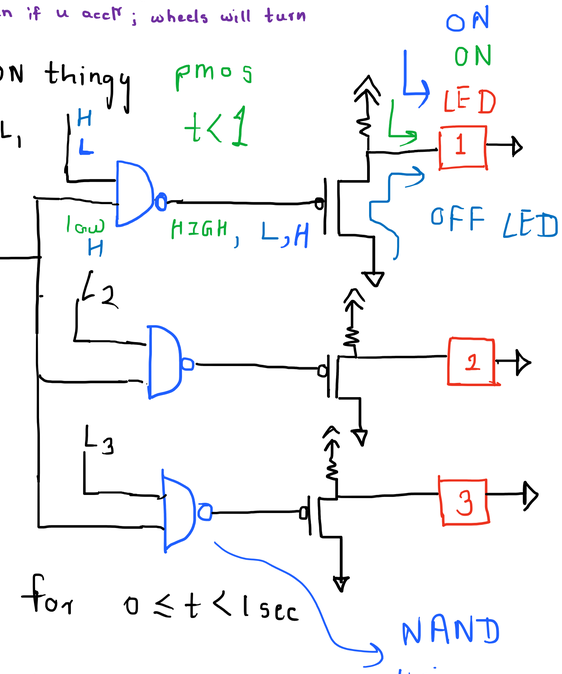



$t < 1$
$V_o$ low

$t > 1$
$V_o$ HIGH

$V_c = V(1 - e^{-t/\tau})$

$\tau$ s.t. when $t = 1sec$
$V_c > V_{ref}$ & $V_o$ HIGH




for $0 \le t < 1sec$

IF $L_1$ is High → No error

vice versa krna hai to just add inverter after $L_1, L_2, L_3$

LOW → ERROR